STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-SMOOTHING-CURVE-ISOLATED-LCI

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3415-3426 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 3428-3452, the complete author proof (23 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 20 of queue_examples-defos.json (pb=1198, stmt 3415-3426, proof 3428-3452).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00SD: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3452–3452. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00SF: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3452–3452. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E7Y: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3329–3336. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BXW: sources/stacks-examples-defos/pinned-original-source.tex, source lines 3452–3452. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let k be a field and let X be a scheme over k. Assume*

(1) *X is separated, finite type over k and $\dim(X) \leq 1$,*
(2) *X is a local complete intersection over k, and*
(3) *$X \to \mathrm{Spec}(k)$ is smooth except at finitely many points.*

Then there exists a flat, separated, finite type morphism $Y \to \mathrm{Spec}(k[[t]])$ whose generic fibre is smooth and whose special fibre is isomorphic to X.

Date: stacks-project:examples-defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. If X is reduced, then we can choose an embedding $X \subset \overline{X}$ as in Varieties, Lemma 0BXW. Writing $X = \overline{X} \setminus \{x_1, \ldots, x_n\}$ we see that $\mathcal{O}_{\overline{X}, x_i}$ is a discrete valuation ring and hence in particular a local complete intersection (Algebra, Definition 00SD). Thus $\overline{X}$ is a local complete intersection over k because this holds over the open X and at the points x_i by Algebra, Lemma 00SF. Thus we may apply Lemma 0E7Y to find a projective flat morphism $\overline{Y} \to \mathrm{Spec}(k[[t]])$ whose generic fibre is smooth and whose special fibre is $\overline{X}$. Then we remove $x_1, \ldots, x_n$ from $\overline{Y}$ to obtain Y.

In the general case, write $X = X' \amalg X''$ where with $\dim(X') = 0$ and X'' equidimensional of dimension 1. Then X'' is reduced and the first paragraph applies to it. On the other hand, X' can be dealt with as in the proof of Lemma 0E7Y. Some details omitted. $\square$